BC3: Portfolio Replica

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Problem and data

Goal. Replicate a hidden "Monster Index" using only liquid Bloomberg futures, subject to a regulatory VaR cap.

$$r_t^{\text{target}} = 0.50\, r_t^{\text{HFRX}} + 0.25\, r_t^{\text{MSCI World}} + 0.25\, r_t^{\text{Global Bond}}$$

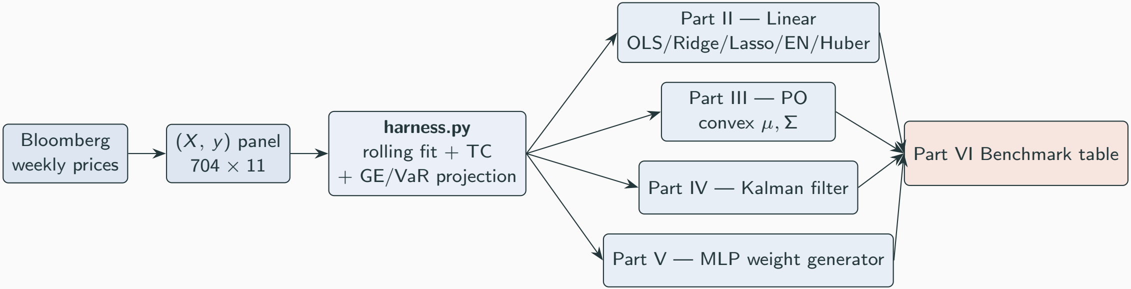
Universe (11 futures).

- Rates: RX1, TY1, DU1, TU2
- Equity: ES1, NQ1, VG1, TP1, LLL1
- Commodity: GC1, CO1

Constraints.

- Sample: weekly, Oct 2007 – Apr 2021 (704 obs).
- Gaussian VaR cap (1%, 1-month) $\leq$ 8%.
- Gross-exposure cap: $\sum_j |w_j| \leq L$.
- Transaction cost: $\tau = 5$ bps on L1 turnover.

Pipeline overview — one harness, four model families



Backtest methodology — Part I

Rolling walk-forward. At every rebalance week t:

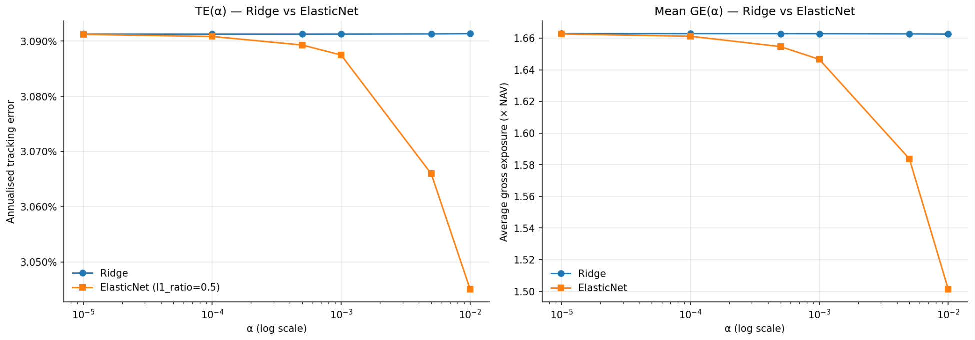
1. **Fit** on $X[t{-}104 : t]$, $y[t{-}104 : t]$ (no look-ahead — $X[t]$, $y[t]$ never seen).
2. **Project** onto the feasible set: gross cap $\sum_j |w_j| \le L$, then iterative shrinkage until $\mathrm{VaR}(w) \le 8\%$.
3. **Trade** $\Delta w_t = w^{\text{new}} - w^{\text{old}}$; **cost** $c_t = \tau \cdot \sum_j |\Delta w_{t,j}|$.
4. **Realise** $r_t^{\text{rep}} = w_t^\top r_t$; on non-rebalance weeks carry w_{t-1} forward.

Contract metrics.

$$\mathrm{TE} = \sqrt{52\,\mathrm{Var}(r_t^{\text{rep}} - y_t)}, \quad \mathrm{IR} = \frac{52\,\mathbb{E}[r_t^{\text{rep}} - y_t]}{\mathrm{TE}}, \quad \mathrm{net\_IR} = \frac{52\,\mathbb{E}[r_t^{\text{rep}} - c_t - y_t]}{\mathrm{net\_TE}}$$

VaR fallback. Jarque-Bera on the target rejects normality ($p \approx 0$); Gaussian VaR (4.14%) understates the historical 1%-quantile (5.96%) — the harness can fall back to a historical VaR projector.

Part II — classical linear benchmark



Ridge vs. ElasticNet across matched α grid: TE flat in α; ElasticNet's average gross exposure collapses once L1 starts zeroing coefficients at $\alpha \geq 10^{-3}$.

Sweep. {OLS, Ridge, Lasso, EN, Huber} $\times$ {all-11, top-5} — single harness: 104-week window, $\Delta t = 4$, scale-only, GE cap 2, VaR cap 20%.

Reading.

- Ridge / EN collapse to OLS at $\alpha < 10^{-3}$.
- Top-5 pruning saves $\sim 4\times$ on average GE for a small TE premium.

Part III — portfolio constraints (post-hoc projection)

Why projection matters. Gaussian VaR is rejected by Jarque-Bera; the historical 1%-quantile is fatter than $\mathcal{N}$. The harness projects w onto a *historical* VaR feasible set.

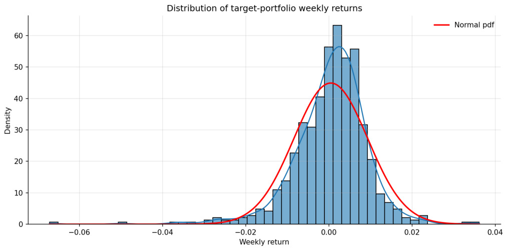
Six base configurations (B1–B6):
We test several portfolio replication configurations by combining different rebalancing frequencies and leverage constraints.

Rebalancing frequencies:

$$\{Weekly, Monthly, Yearly\}$$

Leverage constraints:

$$\{1.0, 2.0\}$$

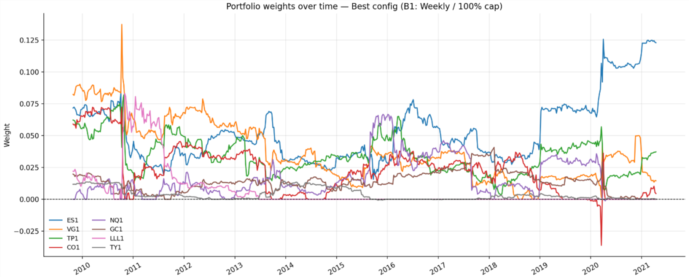


Config	Rebalancing Frequency	Maximum Gross Exposure
B1	Weekly (1w)	100%
B2	Weekly (1w)	200%
B3	Monthly (4w)	100%
B4	Monthly (4w)	200%
B5	Quarterly (12w)	100%
B6	Quarterly (12w)	200%

Part III — results of the different scenarios

Config	Description	Rebal (w)	Max GE	Ann. Return	Ann. Vol	Sharpe	TE	IR	ρ	Mean GE	VaR (1m,1%)	Turnover	Net IR	Net TE	Max DD
B1	Weekly / 100% cap	1	100%	+1.92%	2.91%	0.66	3.91%	-0.461	0.765	19.1%	1.75%	0.0112	-0.468	3.91%	4.27%
B2	Weekly / 200% cap	1	200%	+1.92%	2.91%	0.66	3.91%	-0.461	0.765	19.1%	1.75%	0.0112	-0.468	3.91%	4.27%
B3	Monthly / 100% cap	4	100%	+1.84%	3.03%	0.61	3.91%	-0.480	0.754	19.2%	1.61%	0.0064	-0.484	3.91%	4.52%
B4	Monthly / 200% cap	4	200%	+1.84%	3.03%	0.61	3.91%	-0.480	0.754	19.2%	1.61%	0.0064	-0.484	3.91%	4.52%
B5	Quarterly / 100% cap	12	100%	+1.94%	2.94%	0.66	3.82%	-0.467	0.783	19.4%	1.58%	0.0042	-0.470	3.82%	4.50%
B6	Quarterly / 200% cap	12	200%	+1.94%	2.94%	0.66	3.82%	-0.467	0.783	19.4%	1.58%	0.0042	-0.470	3.82%	4.50%

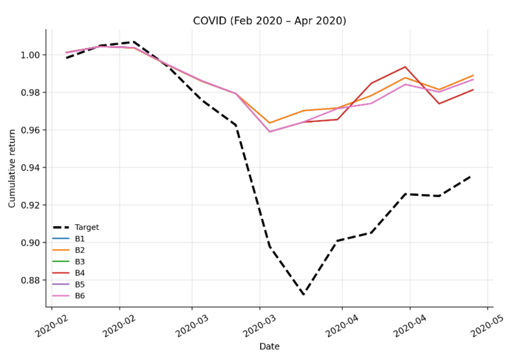
The best configuration is chosen based on the IR metric



Part III — Crisis evolution

Crisis-window evaluation

- COVID period: 2020-02 to 2020-04.
- Used as a stress-test window for the replication strategy.
- All configurations improve the maximum drawdown relative to the target.
- Quarterly rebalancing is the only cadence that keeps tracking error bounded.

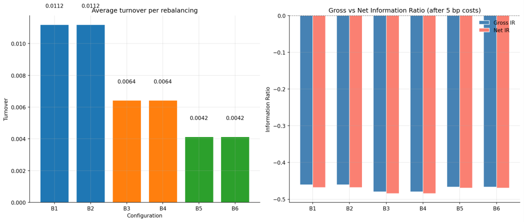


Selection logic

Several configurations are compared using both performance and risk criteria.

- Gross IR measures performance before transaction costs.
- Net IR measures performance after transaction costs.
- Turnover captures the intensity of portfolio rebalancing.
- Tracking error measures how closely the replica follows the target.

The final recommended configuration is selected as the one with the highest **Net IR**.



Turnover and Gross vs Net IR across configurations.

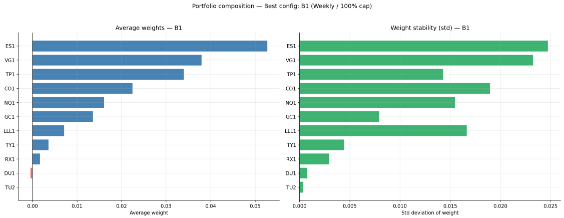
Part III — Recommended portfolio composition

Recommended configuration

The selected configuration is the one maximizing the **Net Information Ratio**.

The portfolio composition is then analyzed through:

- the average weight of each asset;
- the stability of each weight over time



Average weights and weight stability for the recommended configuration.

Part III — predict-then-optimize (convex layer)

Two-stage rebalance.

1. **Alpha.** Fit a classical linear model on $(X_{\text{tr}}, y_{\text{tr}})$; take $\mu = \texttt{coef\_}$.
2. **Risk.** Ledoit–Wolf shrunken sample covariance Σ (annualised).
3. **Optimise** (CVXPY):

$$\max_{w} \;\; \mu^\top w \; - \; \tfrac{1}{2}\lambda \, w^\top \Sigma \, w \; - \; \tau \, \|w - w_{t-1}\|_1 \quad \text{s.t.} \quad \|w\|_1 \le L, \;\; |w_j| \le 0.5$$

Sanity check. With $\lambda = 2$, slack constraints, no LW shrinkage, the convex layer recovers OLS to within solver tolerance.

Findings.

- Longest rolling window (3y) and modest Ridge dominate net IR on the leaderboard.
- Huber is competitive only in the 2008/2020 windows.
- Lasso's sparsity buys lower turnover but loses on TE.

Part IV — Kalman filter / state-space formulation

$$
\begin{aligned}
x_t &= A\,x_{t-1} + B\,u_t, && u_t \sim \mathcal{N}(0, I_{11}) && \text{(state: portfolio weights, random walk)}\\
y_t &= C_t\,x_t + D\,\varepsilon_t, && \varepsilon_t \sim \mathcal{N}(0, 1) && \text{(observation: target return)}
\end{aligned}
$$

$A = I_{11}$, $B = \sigma_w \cdot I_{11}$, $C_t = r_t^\top$, $D = \hat{\sigma}_y$ from the training window, $x_t \in \mathbb{R}^{11}$ the latent weight vector, y_t the scalar target return. Ridge regression warm-start on the first 52 weeks; filter implemented from scratch.

Three variants:

- **Basic Kalman** — fixed process noise $\sigma_w = 0.001$, fixed R, with a VaR guardrail.
- **Tuned Kalman** — grid search over $(Q, R_{\text{scale}}, P_0)$ with a Universal Replication Score, plus Q-sensitivity analysis.
- **Double Adaptive Volatility** — both Q and R scale dynamically with rolling volatility.

Parameter Robustness. $\pm 20\%$ perturbation grid confirms structural soundness.

Look-ahead discipline. $\hat{x}_{t|t}$ uses y_t (unavailable at the open of week t). We shift weights by one week: held weight at t depends only on observations through $t-1$.

Part IV — Kalman constraints & optimisation score

1. Turnover Dampening

$$\hat{w}_t^{\text{damp}} = (1-\alpha)\,\hat{w}_{t-1}^{\text{final}} + \alpha\,\hat{x}_{t|t}$$

Exponential smoothing on raw Kalman updates, structurally slowing weight adjustments to prevent excessive trading and reduce transaction costs.

2. Gross Exposure Cap

$$\hat{w}_t^{\text{gross}} = \begin{cases} \hat{w}_t^{\text{damp}} \dfrac{GE_{\text{cap}}}{\sum_i |\hat{w}_{i,t}^{\text{damp}}|} & \sum_i |\hat{w}_{i,t}^{\text{damp}}| > GE_{\text{cap}} \\ \hat{w}_t^{\text{damp}} & \text{otherwise} \end{cases}$$

Proportional scale-down if the absolute sum of weights exceeds $GE_{\text{cap}} = 2.0$.

3. VaR Guardrail

$$\hat{w}_t^{\text{final}} = \begin{cases} \hat{w}_t^{\text{gross}} \dfrac{VaR_{\text{cap}}}{\widehat{VaR}} & \widehat{VaR} > 8\% \\ \hat{w}_t^{\text{gross}} & \text{otherwise} \end{cases}$$

where $\widehat{VaR} = VaR_{1\%,\,4W}(\hat{w}_t^{\text{gross}})$. Aggressively downweights if forward risk breaches 8%.

4. Replication Score

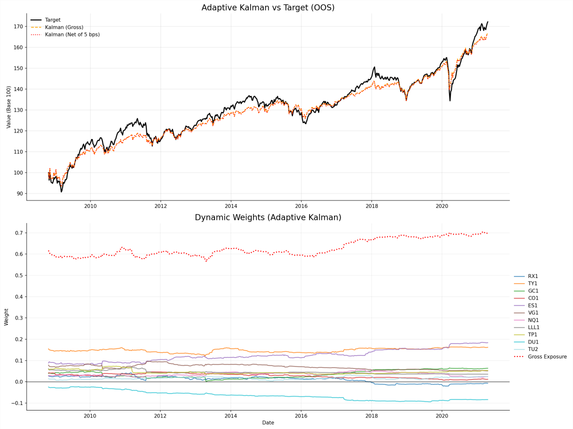
$$S = \rho(\mathbf{r}_R, \mathbf{r}_T) - 2\,TE_n - 0.5\,\overline{\Delta w} - 0.1\,MDD_n$$

TE_n = net TE, $\overline{\Delta w}$ = avg turnover, MDD_n = net max drawdown.

Part IV — Adaptive Kalman: results comparison

Metric	Basic	Adaptive
net_IR	−0.141	**−0.089**
IR	−0.137	**−0.083**
ρ	0.868	**0.879**
TE (net)	3.04%	**2.89%**
GE (avg)	0.59	0.62
$\overline{T}$	0.005	0.007
MDD (net)	14.2%	**12.8%**

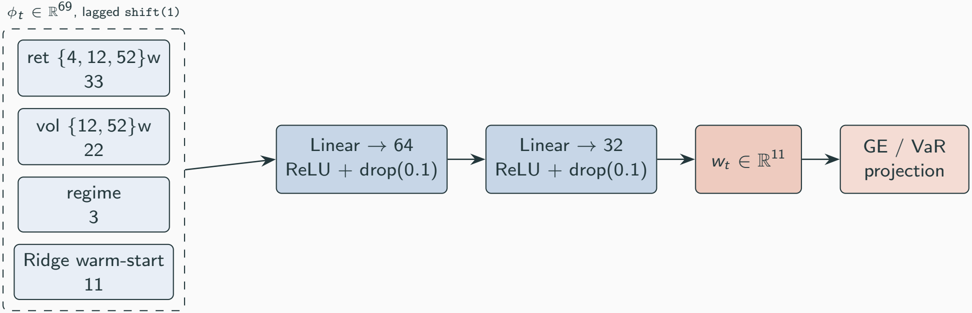
The adaptive variant improves across all risk-adjusted metrics: higher ρ, lower TE, smaller drawdown.



Adaptive Kalman replica vs. target cumulative returns.

Part V — deep learning weight generator (architecture)

End-to-end. The network outputs the *weights*: $w_t = f_\theta(\phi_t)$, $r_t^{\text{rep}} = w_t^\top r_t$.



Loss — forward-window MSE + turnover + budget:

$$\mathcal{L}(\theta) = \underbrace{\mathrm{MSE}_{s\in[t,\,t+H-1]}\big(w_t^\top r_s - y_s\big)}_{H=12} + \lambda_1 \|\Delta w\|_1 + \lambda_2 \|\Delta w\|_2^2 + \gamma\,(1^\top w_t - 1)^2$$

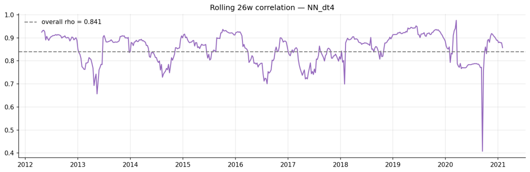
Part VI — consolidated leaderboard (top of 30 rows)

Model	net_IR	IR	ρ	TE	GE	$\bar{T}$
NN $\Delta t = 4$	**+0.170**	+0.177	0.841	2.96%	1.02	0.008
NN $\Delta t = 1$	+0.135	+0.143	0.843	2.93%	1.03	0.010
M3-PO | Huber $\delta = 1.35$ | all-11	+0.017	+0.032	0.856	2.93%	1.28	0.017
M3-PO | OLS | all-11	+0.017	+0.032	0.853	2.95%	1.38	0.018
M2 | Huber | all-11	-0.086	-0.022	0.844	3.03%	1.59	0.075
Kalman (basic, $\sigma_w = 10^{-3}$)	-0.141	-0.137	0.868	3.04%	0.59	0.005
M2 | Lasso / EN | top-5	-0.185	-0.173	0.828	3.18%	0.37	0.014
M3-Cstr | B1 (weekly / 100% cap)	-0.468	-0.461	0.765	3.91%	0.19	0.011

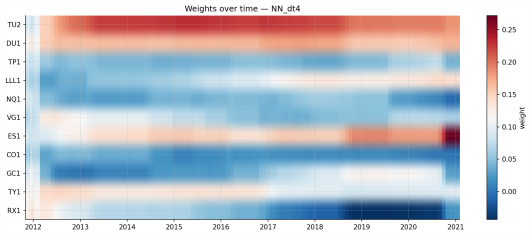
NN $\Delta t = 4$ tops the leaderboard on net IR with $\rho = 0.84$ at the lowest gross exposure and turnover of the positive-IR group.

Best model spotlight — cumulative & weights

NN $\Delta t = 4$ vs target, OOS 2011-11 $\rightarrow$ 2021-02.



Rolling 26-week correlation; overall $\rho = 0.84$.



Asset weights over time (RdBu = long/short).

Reading the heatmap.

- Persistent long in rates (RX1, TY1) carries the bond leg.
- Equity loadings (ES1, NQ1, VG1) rotate with the regime.
- Sign-flips are mild — the network respects the turnover penalty.